

PRATEESH PATLOLLA

Data Scientist / ML Engineer

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SUMMARY

Data Scientist with 3+ years of experience delivering GenAI, NLP, Forecasting, and Optimization solutions at scale. Built cloud-native ML systems using Databricks and AWS, Azure, Spark, Databricks, and MLOps tools from experimentation to deployment. Delivered high-impact products including LLM-powered RAG assistants (LangChain, FAISS, LangGraph, OpenAgents, Bedrock), Gurobi-based production optimizers, and real-time APIs for cross-functional teams. Skilled in A/B testing, Causal inference, and Stakeholder-facing experimentation, with a track record of driving measurable ROI across enterprise environments.

SKILLS

- **Languages:** Python, SQL, R, Java, JavaScript
- **Big Data & Pipelines:** Apache Spark, PySpark, Hadoop, Hive, Parquet, Kinesis Firehose, Glue
- **Cloud & MLOps:** Azure (Azure Databricks, Monitor, OpenAI), AWS (SageMaker, Bedrock, Redshift, Lambda, CloudWatch), Docker, Kubernetes, Airflow, Jenkins, FastAPI, GitHub Actions, VPC Networking
- **GenAI & LLMs:** LangChain, LangGraph, OpenAgents, Guardrails AI, Llama 2, GPT-4, Prompt Engineering, Vector DBs (FAISS, Pinecone, Chroma), LLMOps tools (PromptLayer, TruLens)
- **Databases & Storage:** Redis (ElastiCache), DynamoDB, PostgreSQL, Oracle, Redshift, MongoDB
- **Analytics & BI:** Tableau, Power BI, AWS QuickSight
- **Modeling & Optimization:** Scikit-learn, XGBoost, SARIMAX, PyTorch, SHAP, Gurobi, A/B Testing, Causal Inference
- **Monitoring:** Azure Monitor, CloudWatch, Datadog

PROFESSIONAL EXPERIENCE

Apple
Data Scientist

Austin, TX
Nov 2025 – present

- Engineered a semi-automated evaluation pipeline using fine-tuned Google Gemini models and active learning strategies, reducing human annotation dependency by 70% while maintaining label fidelity.
- Developed an automated capacity planning workflow integrating Quip and Slack APIs; built a preprocessing pipeline that detects data anomalies and triggers automated Slackbot notifications, ensuring data readiness for budget and capacity planning.
- Automated the generation of comparative experimentation reports for App Store Search PLB2@7 ranking metrics into Keynote presentations, streamlining cross-functional decision-making for search ranking algorithms.
- Designed intelligent task routing heuristics and workforce capacity models to forecast seasonal demand, optimizing annotator-to-task allocation to maximize throughput and quality.

Toyota North America
Data Scientist

Plano, TX
Feb 2023 – Nov 2025

- Led the design and POC of AskToyota, a GenAI chatbot using LangChain, FAISS, LangGraph, and LLaMA 2, deployed in secure VPC environments with IAM-based access control and TLS. Reduced stakeholder turnaround time by 30% by surfacing forecasts and KPIs on demand.
- Spearheaded Accessory Recommendation engine generating \$150M+ annual revenue using XGBoost, dynamic segmentation, and incentive-aware retraining, integrated A/B testing, and causal inference for robust evaluation.
- Built Toyota's Annual Production Optimizer (APO) using Gurobi to solve multi-objective LP problems (profit, volume, GHG score) across 14 global factories, optimizing \$700M+ in planning annually.
- Developed Smart Forecasting engine with PySpark and SARIMAX to predict dealer-level sales; enabled 3+ hours/month time savings per dealer and built self-serve analytics dashboards via Tableau.
- Productionized Pyspark and ML pipeline in Azure Databricks, integrating model registry, drift detection with auto-retraining, Airflow orchestration, and CloudWatch/Datadog monitoring for real-time alerts.
- Deployed real-time ETA API for dealer websites using ElastiCache Redis with SHA-1 for online inference; batch-trained models using PySpark and Parquet on S3. Packaged with Docker and deployed via Kubernetes.
- Blended causal inference and A/B testing frameworks with GenAI tools (Azure OpenAI, LangChain agents, Guardrails AI, Azure Cognitive Search, FAISS) on Azure Databricks to auto-explain accessory model KPIs, increasing dealer conversions by 20% and boosting stakeholder trust..

Amazon, Alexa AI
Data Scientist Intern

Santa Barbara, CA
May 2022 – August 2022

- Retrained and deployed Alexa's question categorizer using advanced NLP (multi-class classification, intent detection), improving query routing across 100+ categories via SageMaker.
- Used resampling and confidence intervals to evaluate and improve model confidence on rare query intents, increasing trust and accuracy for low-frequency categories in Alexa.
- Improved the generated success rate model's alignment with human-labeled success rate, reducing gap from ~20% to under 14%, enabling reduced annotation spend.
- Built reporting pipelines that pushed KPIs into Redshift and Quicksight to visualize key health metrics and rolling forecast 3-month trends to support weekly reviews with product stakeholders.

EDUCATION

INDIANA UNIVERSITY BLOOMINGTON
Masters of Science in Data Science

Bloomington, Indiana, USA

GITAM UNIVERSITY
Bachelor's in Computer Science

Hyderabad, India

PROJECTS

Prateesh's Gen AI Resume Assitant

link: prateeshreddy.github.io

- Developed a scalable Retrieval-Augmented Generation (RAG) pipeline using LangChain, FAISS, and OpenAI's embeddings, enabling context-aware, accurate Q&A about my professional background.
- Engineered advanced prompt templates and conversational memory, delivering enterprise-grade efficiency with 98% retrieval accuracy, 120 ms latency (p99), and inference costs below \$0.0005 per query.
- Deployed serverlessly on Cloudflare Workers with robust CI/CD via GitHub Actions, featuring real-time observability and automated logging for iterative improvements.

Cognitive Search Engine by NLP

- Provided Cognitive search capability for Eli Lilly and Company to search against databases like FDA and EMA via natural language questions and return relevant results to help with accelerating regulatory submissions for Eli Lilly as an Intern in the Summer of 2021 using Hugging Face transformers models.

Research Scientist at Institutional Analytics, Indiana University

- Been part of the guest lecture for ILSZ 637 - Information Visualization by Noriko Hara in Spring 2022, a graduate-level course talking about data storytelling and data visualization.
 - Built data pipelines with Tableau reports to analyze student performance, retention, and graduation trends, influencing key academic decisions.
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